

Given:

To build a Logistic Regression model that classifies breast tumors as malignant or benign using the Breast Cancer dataset and evaluate its performance using the confusion matrix, precision, recall, ROC-AUC score, and threshold tuning.

DATASET: [Click to Download Breast Cancer Wisconsin \(Diagnostic\) Data Set](#)

LogisticRegression.py

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    confusion_matrix,
    ConfusionMatrixDisplay,
    precision_score,
    recall_score,
    roc_auc_score,
    roc_curve
)
df = pd.read_csv(r"..\\DataSets\\data.csv")
df = df.drop("id", axis=1)
df["diagnosis"] = df["diagnosis"].map({"M": 1, "B": 0})
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]
print("Confusion Matrix")
print(confusion_matrix(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("ROC-AUC Score:", roc_auc_score(y_test, y_prob))
cm = confusion_matrix(y_test, y_pred)
ConfusionMatrixDisplay(cm).plot()
plt.title("Confusion Matrix")
plt.show()
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
plt.figure(figsize=(8,6))
plt.plot(fpr, tpr, label="ROC Curve")
```

```

plt.plot([0,1],[0,1],'r--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
threshold = 0.3
y_pred_threshold = (y_prob >= threshold).astype(int)
print("\nThreshold:", threshold)
print("Precision:", precision_score(y_test, y_pred_threshold))
print("Recall:", recall_score(y_test, y_pred_threshold))

```

Output:

Confusion Matrix

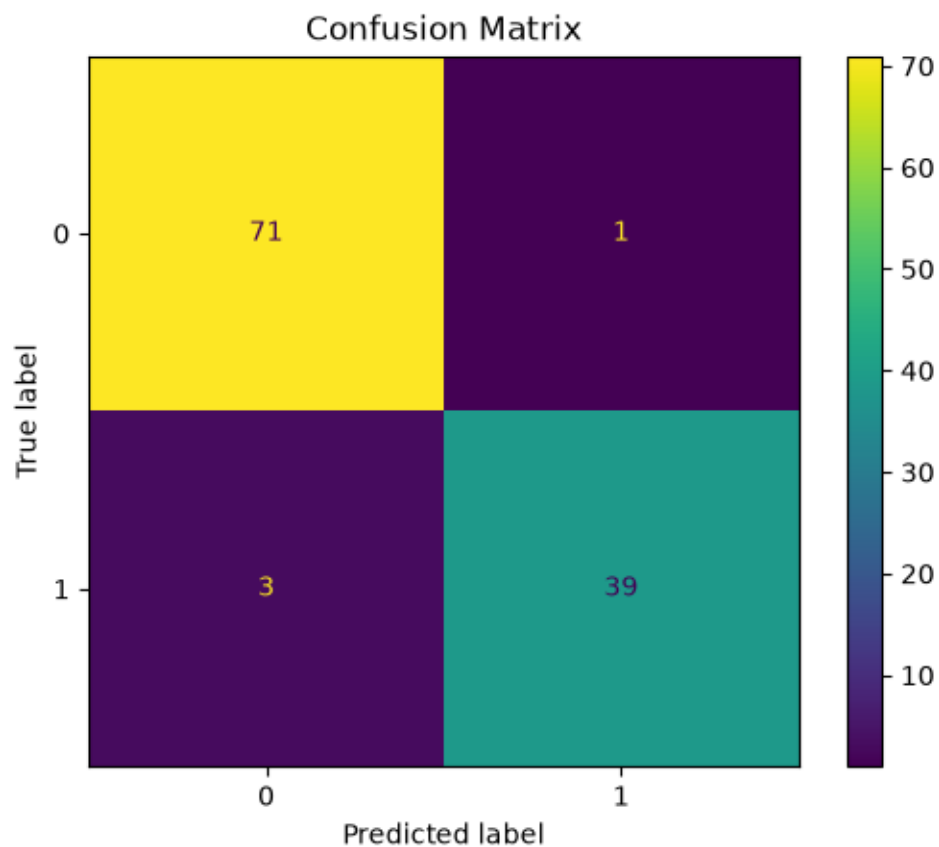
```
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```

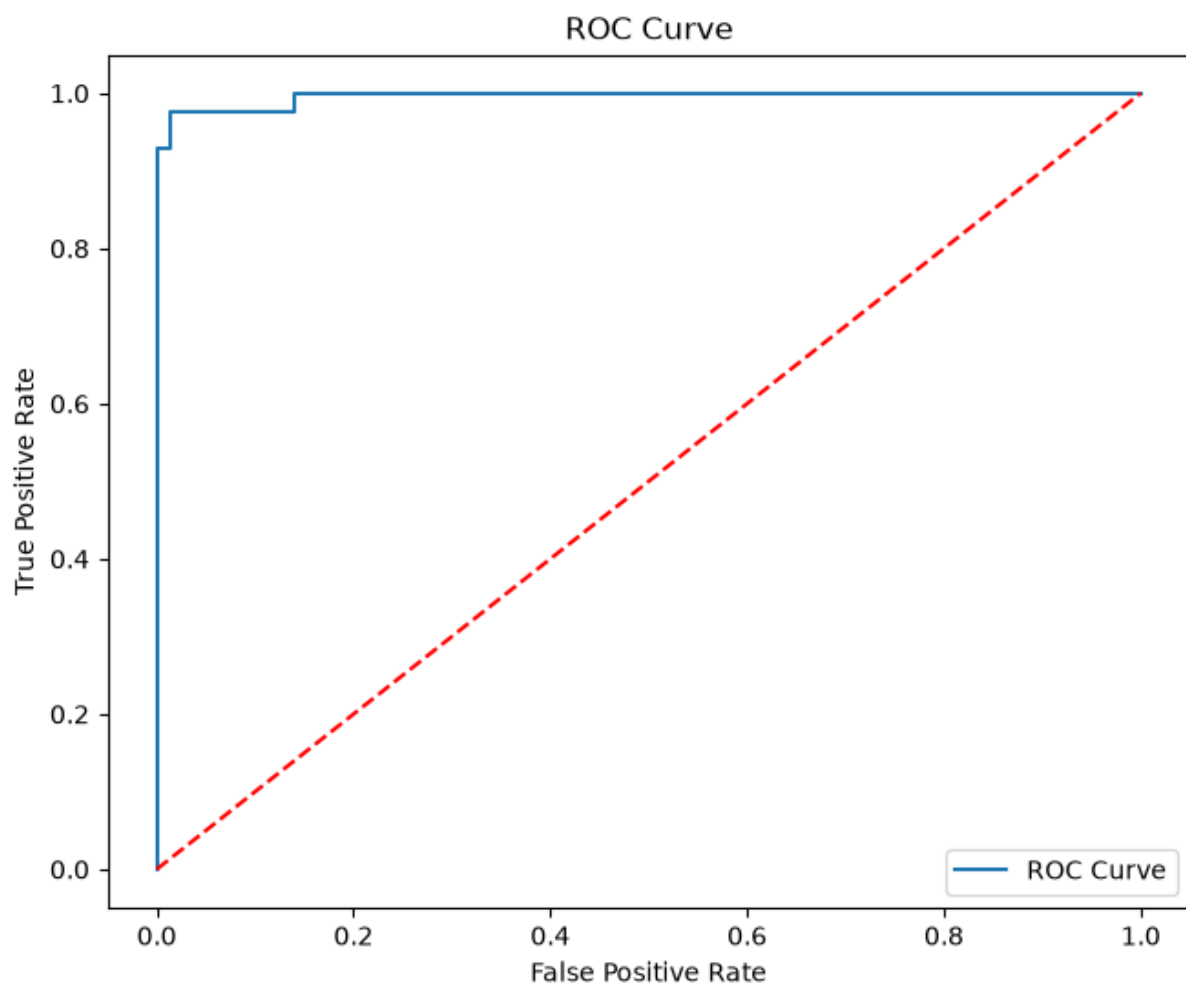
```
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```

Precision: 0.975

Recall: 0.9285714285714286

ROC-AUC Score: 0.996031746031746





Threshold: 0.3

Precision: 0.9761904761904762

Recall: 0.9761904761904762